A PROJECT TITLE
ON
WATER QUALITY CLASSIFICATION USING SUPPORT VECTOR MACHINE AND RANDOM FOREST

project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING
BACHELOR OF TECHNOLOGY
In
A.Y.: 2026-27

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Abstract

This project develops a binary machine-learning classification system for water samples using 20 numerical physicochemical and biological parameters. The submitted project archive contains 1,000 raw records and 21 columns, including the target variable is_safe. Inspection found 500 exact duplicate rows and no missing values. After duplicate removal, 500 unique records remain, with 327 samples labelled SAFE (1) and 173 labelled UNSAFE (0). The documented pipeline uses an 80:20 stratified train-test split with random_state=42. StandardScaler is fitted only on the training data for the SVM, while Random Forest uses the original numerical feature values.

Keywords: Water Quality Classification; Support Vector Machine; Random Forest; Data Preprocessing; Flask; Machine Learning

1. Introduction

1.1 Purpose of the Project

Water quality assessment involves multiple chemical and biological measurements, so classification based on a single parameter is not sufficient for this project. The implemented system uses 20 numerical parameters—aluminium, ammonia, arsenic, barium, cadmium, chloramine, chromium, copper, flouride, bacteria, viruses, lead, nitrates, nitrites, mercury, perchlorate, radium, selenium, silver, and uranium—to learn patterns associated with the target label is_safe.

The motivation is to provide a quick exploratory classification from a multivariate sample. The project includes a Flask-based web application with a browser frontend, model prediction endpoints, and SQLite storage for prediction history. The application explicitly treats its outputs as statistical estimates and not as a substitute for certified laboratory testing.

1.2 Problem Statement

The problem is formulated as a binary supervised classification task. Given numerical measurements for the 20 water-quality parameters, the system predicts the value of is_safe, where class 0 is labelled UNSAFE and class 1 is labelled SAFE in the project. The expected output is a binary classification from the trained SVM and Random Forest models, with classifier probability estimates also available in the prediction interface.

1.3 Existing Methods and Disadvantages

Conventional water assessment can involve laboratory measurement and comparison of individual parameters with established limits. Such approaches remain important for formal water-quality decisions, but they do not by themselves provide a learned multivariate classification model. The submitted project does not contain a separate conventional baseline experiment, so no numerical performance disadvantage is claimed here.

From the machine-learning perspective, model performance can be affected by duplicate records, scaling, class distribution, parameter choices, and the particular train-test split. This project addresses several of these factors by removing exact duplicates, using stratification, applying training-only StandardScaler for SVM, and comparing two different classifier families. The archive does not include a systematic hyperparameter-search experiment.

1.4 Proposed System

The proposed system follows a reproducible pipeline: inspect the dataset; remove exact duplicates; verify missing values and the target label; perform exploratory analysis; separate features and target; use an 80:20 stratified train-test split; fit StandardScaler only on training data for SVM; train an RBF-kernel SVM and a 200-tree Random

Forest; evaluate accuracy, precision, recall and F1-score; compare the models; and expose the saved models through a Flask prediction interface with SQLite prediction history.

1.5 Research Objectives(atleast 4)

1. Prepare and inspect the water-quality dataset and identify the target variable.
2. Clean the dataset by removing duplicate records and verifying missing values.
3. Perform exploratory analysis of the 20 water-quality parameters and target distribution.
4. Prepare a stratified train-test split and apply leakage-safe feature scaling for SVM.
5. Train Support Vector Machine and Random Forest classification models using the prepared data.
6. Evaluate and compare the models using accuracy, precision, recall and F1-score.

2. Literature Survey (recent studies from 2-3 years)

Review ten recent and directly relevant studies. Verify bibliographic details, DOI, volume, issue, pages/article number, and year before final submission.

S.No.	Title	Reference with DOI	Problem or gap addressed	Objective of the Research	Focus of the paper (Methods used)	Inference	Recommendations for future research
1	Predicting water potability using a machine learning approach	El-Bacha, R., Salhi, A., Hafid, A., & Rabi, S. (2025), Environmental Challenges, 19, 101131. DOI: 10.1016/j.envc.2025.101131	Rapid potability assessment is difficult when many physical and chemical variables must be considered.	Evaluate ML models for water-potability prediction.	SVM and Random Forest using a public water-potability dataset.	Random Forest outperformed SVM in the reported study; RF accuracy was 70%.	Test larger, geographically diverse datasets and independent validation.
2	Water Quality Assessment Using the Random Forest Classification Model	Bouchraki, F., Hamchaoui, S., Ayad, L. L., Fetouh, Y., Mezhoud, C., & Berreksi, A. (2025), Water Environment Research, 97(10), e70197. DOI: 10.1002/wer.70197	Automated water-quality classification needs to handle real and synthetic records while maintaining class balance.	Develop an automated RF classification model and decision-support tool.	Random Forest, stratified cross-validation, mixed real/synthetic data, and a web platform.	Average macro F1 was 0.98; the real-data test set achieved 99% accuracy.	Improve balancing and validation with more real-world samples.
3	Advanced machine learning models for accurate water quality classification and WQI prediction: Implications for aquatic disease risk management	Hridoy, M. A. A. M., Shawkat, A. I., Bordin, C., Acharjee, M. R., Masood, A., Baki, A. O., & Mamun, M. A. (2025), Science of the Total Environment, 1008, 180965. DOI: 10.1016/j.scitotenv.2025.180965	Water-quality classification and WQI prediction require accurate models and useful predictors.	Compare tuned ML models for classification and WQI prediction.	LightGBM, XGBoost, Random Forest and SVM with grid-search optimization.	The study reports very high classification performance and highlights important water-quality variables.	Validate the tuned framework on additional regions and real-time data.
4	AQuA-P: A machine learning-based tool for water quality assessment	Díaz-González, L., Aguilar-Rodríguez, R. A., Pérez-Sansalvador, J. C., & Lakouari, N. (2025), Journal of Contaminant Hydrology, 269, 104498. DOI: 10.1016/j.jconhyd.2025.104498	Water-quality classification across groundwater and surface-water types needs a practical and consistent tool.	Build an ML-based water-quality assessment tool using a national Mexican database.	XGBoost, SVM, KNN, decision trees and multinomial logistic regression.	Decision Trees performed best in the reported comparison and the study produced a user-oriented tool.	Validate across additional regions and maintain broader datasets.
5	Machine learning approaches for predicting water quality towards climate-resilient groundwater management in southern India	Rudrani, G. A., Madhnure, P., Yadav, A., Kumari, B., Roy, A., Kumar, V., Dauji, S., & Tirumalesh, K. (2025), Hydrology Research, 56(8), 754–773. DOI: 10.2166/nh.2025.042	Seasonal and regional variation complicates groundwater-quality prediction.	Predict drinking-water quality index for groundwater in southern India.	Random Forest and multiple linear regression with seasonal evaluation.	The study reports strong prediction performance but notes seasonal constraints for RF.	Test more seasons, locations and model families.
6	A review of machine learning and internet-of-things on the water quality assessment: Methods, applications and future trends	Dharmarathne, G., Abekoon, A. M. S. R., Bogahawaththa, M., Alawatugoda, J., & Meddage, D. P. P. (2025), Results in Engineering, 26, 105182. DOI: 10.1016/j.rineng.2025.105182	Traditional monitoring can be costly, time-intensive and limited for real-time insight.	Review ML and IoT methods, applications and research gaps.	Review of RF, boosting, SVM, neural networks, IoT and explainable AI.	The review identifies growing ML/IoT use and underuse of explainable AI.	Develop interoperable, explainable and scalable real-time systems.

7	Comprehensive analysis of multiple classifiers for enhanced river water quality monitoring with explainable AI	Ramya, S., Srinath, S., & Tuppad, P. (2024), Case Studies in Chemical and Environmental Engineering, 10, 100822. DOI: 10.1016/j.cscee.2024.100822	River-water classification benefits from comparing classifiers and adding interpretability.	Evaluate multiple classifiers for WQI classification and interpretation.	Fourteen classifiers including SVM and RF; confusion matrices, classification reports, ROC curves, LIME and SHAP.	Gradient Boosting achieved the highest reported accuracy; SVM and RF also performed strongly.	Validate explanations and models on additional river systems.
8	Intelligent water quality prediction system with a hybrid CNN–LSTM model	Guo, H., Chen, Z., & Teo, F. Y. (2024), Water Practice & Technology, 19(11), 4538–4555. DOI: 10.2166/wpt.2024.282	Water-quality prediction can benefit from temporal forecasting and classification.	Develop an intelligent system for water-quality prediction and classification.	MLP, XGBoost, LSTM, CNN–LSTM, RF, KNN, SVM and LightGBM.	The hybrid CNN–LSTM performed well for pH/DO forecasting; SVM achieved the highest WQC accuracy at 88.75% in the study.	Test the framework with additional parameters and regions.
9	Machine learning approach for water quality predictions based on multispectral satellite imageries	Anand, V., Oinam, B., & Wieprecht, S. (2024), Ecological Informatics, 84, 102868. DOI: 10.1016/j.ecoinf.2024.102868	Field-based water-quality monitoring can be spatially limited.	Estimate and map water-quality parameters from multispectral satellite data.	SVM, Random Forest and multi-linear regression using Sentinel-2 and ResourceSat-2 imagery.	SVM performed best among the evaluated models for the reported water-quality parameter estimation.	Expand validation across seasons, sensors and water bodies.
10	Machine Learning for Water Quality Index Forecasting	Doddabasappaar, A. K. T., Yogendra, B. E., Janardhan, P., & Siddegowda, P. N. (2024), Emerging Science Innovation, 3, 43–53. DOI: 10.46604/emsi.2024.12870	Increasing pollution levels create a need for timely WQI forecasting.	Forecast WQI using supervised ML methods in Tumkur district, Karnataka.	SVM, regression trees, linear regression and neural networks with an 80:20 split.	SVM and linear regression showed strong reported predictive performance for WQI.	Use larger datasets and test robustness across seasons and locations.

3. Methodology

3.1 Proposed System Flow Diagram

The implemented workflow is:

Dataset → Data inspection → Duplicate removal → Missing-value check → EDA → Feature/target separation → Stratified 80:20 split → StandardScaler for SVM → SVM and Random Forest training → Five-fold stratified cross-validation during training → Test-set evaluation → Model comparison → Flask prediction interface

3.2 Dataset Description

The dataset file water_quality.csv included in the project archive contains 1,000 records and 21 columns: 20 numerical water-quality features and the binary target is_safe. The features are aluminium, ammonia, arsenic, barium, cadmium, chloramine, chromium, copper, flouride, bacteria, viruses, lead, nitrates, nitrites, mercury, perchlorate, radium, selenium, silver, and uranium. The raw dataset contains 500 exact duplicate rows and no missing values. After duplicate removal, 500 unique records remain, comprising 173 UNSAFE samples (34.6%) and 327 SAFE samples (65.4%). The archive does not state the original external provenance of the CSV.

3.3 Experimental Setup

The project uses Python scripts for data loading, cleaning, EDA, preprocessing, model training, evaluation and prediction, together with a Flask REST API, browser frontend, and SQLite database. The backend requirements specify Flask 3.0.3, flask-cors 4.0.1, pandas 2.2.2, scikit-learn 1.7.2, joblib 1.4.2, matplotlib 3.9.2, and streamlit 1.39.0. The archive does not explicitly record the operating system, IDE, or Python interpreter version, so those details are not asserted in this report.

3.4 Data Preprocessing

- Data inspection and schema validation were performed, including shape, column names, data types, missing values, duplicates and target distribution.
- Exact duplicate rows were removed. The raw dataset contains 500 duplicate rows, leaving 500 unique records.
- Missing-value analysis found 0 missing values, so no imputation was required.
- No separate outlier-treatment operation is implemented in the supplied cleaning/preprocessing scripts.
- No categorical variables are present; all 20 predictors are numerical, so categorical encoding is not required.
- StandardScaler is fitted only on training data for SVM; Random Forest receives the unscaled feature values.
- The target is the integer label is_safe with classes 0 (UNSAFE) and 1 (SAFE).
- An 80:20 train-test split is performed with random_state=42 and stratification on the target, preventing class-proportion drift between train and test sets.

3.5 Exploratory Data Analysis

The supplied EDA script examines the cleaned dataset using feature distributions, class balance, box plots, a correlation matrix, and correlations between each predictor and is_safe. It is designed to generate a class-distribution chart, a 5×4 grid of feature histograms, feature box plots, a correlation heatmap, and a target-correlation chart. The submitted website ZIP contains the EDA source script but does not include the generated results/graphs directory, so this report describes the implemented EDA rather than claiming that those image files are present in the archive.

3.6 Feature Engineering and Feature Selection (if applicable)

The project does not implement derived features or an automated feature-selection algorithm. All 20 numerical parameters are retained as input features. Scaling is model-specific preprocessing: SVM uses StandardScaler, while Random Forest is trained on the original numerical feature values.

3.7 Model Building and Training

Two binary classifiers are trained. The SVM uses an RBF kernel with C=1.0, gamma="scale", probability=True, and random_state=42. The Random Forest uses 200 decision trees, max_features="sqrt", random_state=42, and n_jobs=-1. Both training scripts use StratifiedKFold with 5 folds, shuffle=True and random_state=42, and cross_val_score with F1 scoring. The final models are fitted on the training data and serialized as svm_model.pkl and random_forest_model.pkl.

3.8 Hyperparameter Tuning(if applicable)

The supplied training scripts do not perform GridSearchCV or RandomizedSearchCV. They define fixed model parameters and use five-fold stratified cross-validation to report F1 performance during training. Therefore, no separately optimized hyperparameter set is reported for this project.

3.9 Model Evaluation

The evaluation code calculates Accuracy, Precision, Recall and F1-score and creates confusion-matrix plots. Because scikit-learn's default positive label is 1, the reported Precision, Recall and F1-score refer to the SAFE class (is_safe=1). ROC-AUC is not calculated or stored by the supplied evaluation workflow, so it is reported as Not recorded. A reproducible re-evaluation of the submitted saved models is used for the final performance table below.

4. Results and Discussion

4.1 Preprocessing Results

The raw dataset contains 1,000 rows and 21 columns, with 0 missing values and 500 duplicate rows. Duplicate removal produces 500 unique rows while retaining all 21 columns. The cleaned target distribution is 173 UNSAFE samples (34.6%) and 327 SAFE samples (65.4%). The preprocessing pipeline then creates 400 training records and 100 test records using an 80:20 stratified split. Twenty numerical predictors are retained, and StandardScaler is fitted only on the training data for SVM.

4.2 Model Performance

Model	Accuracy	Precision	Recall	F1-score	ROC AUC
SVM	92.00%	92.54%	95.38%	93.94%	Not recorded
Random Forest	92.00%	93.85%	93.85%	93.85%	Not recorded
Not applicable	—	—	—	—	Not recorded
Not applicable	—	—	—	—	Not recorded
Not applicable	—	—	—	—	Not recorded

The performance values below were obtained by re-evaluating the submitted serialized SVM and Random Forest models on the cleaned 500-row dataset using the documented 80:20 stratified split with random_state=42. SVM achieved 92.00% accuracy, 92.54% precision, 95.38% recall and 93.94% F1-score. Random Forest achieved 92.00% accuracy, 93.85% precision, 93.85% recall and 93.85% F1-score. These values are the reproducible test-set results of the model artifacts included in the submitted ZIP.

4.3 Visual Results and discussion

The supplied EDA and evaluation scripts are designed to generate class-distribution, feature-distribution, box-plot, correlation, target-correlation, and confusion-matrix graphs, together with a model-comparison graph. The submitted website ZIP does not include the generated results/graphs files, so no figure is inserted here. The class distribution is 327 SAFE and 173 UNSAFE after duplicate removal.

An important artifact inconsistency was found during verification: data/model_results.csv in the submitted ZIP contains SVM values of 96.00% accuracy and 97.01% F1-score and Random Forest values of 97.50% accuracy and 98.11% F1-score. Those archived values do not reproduce when the serialized .pkl models in the same ZIP are evaluated on the documented cleaned dataset and split. Therefore, the final report uses the reproducible 92.00% test accuracy results above rather than presenting the inconsistent archived CSV values as verified model results.

5. Conclusion

The project implements a binary water-quality classification pipeline using 20 numerical water-quality parameters. The raw dataset is inspected, exact duplicates are removed, missing values are verified, EDA is performed, and the cleaned data are split using stratification. SVM is trained with an RBF kernel after training-only standardization, while Random Forest is trained on the original numerical features. Re-evaluation of the submitted serialized models on the documented test split gives 92.00% accuracy for both models; SVM has a slightly higher F1-score (93.94%) than Random Forest (93.85%), while Random Forest has higher precision (93.85% versus 92.54%). The project also integrates the models into a Flask-based web application with SQLite prediction history. The application treats predictions as estimates and not as a replacement for certified laboratory testing.

6. Future Work

- Expand the dataset with more geographically and temporally diverse water samples and document the original data provenance.
- Validate the stored models on an independent external dataset before operational use.
- Synchronize data/model_results.csv and the website metrics with the exact serialized models and documented evaluation split.
- Perform systematic hyperparameter optimization using grid or randomized search with cross-validation.
- Add explainable-AI methods such as SHAP or LIME to help interpret model predictions.
- Investigate class-imbalance handling and robustness under changing water-quality distributions.
- Add ROC-AUC and other threshold-based evaluation outputs to the archived evaluation results.

7. References

References are presented in a consistent APA-style format with DOI information checked against publisher, PubMed, DOAJ, or article pages.

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Signature of the Guide

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